Lewis Chase

Manufacturing Engineer | Controls & Automation

(786) 894-3151 • Online-Resume • lewis.chase@protonmail.com

PROFESSIONAL SUMMARY

Reliable and detail-oriented engineering student with over five years of hands-on experience in industrial, manufacturing, and automation environments. Proficient in assembly line design, automation systems, and process optimization. Adept at PLC and HMI programming, troubleshooting, and integrating advanced technologies, including robotics and vision systems.

WORK EXPERIENCE

Wegmann Automotive - *Murfreesboro, TN*
Process Engineer *Nov 2023 - Present*

- Interpreted and applied electrical and mechanical design drawings to develop solutions.
- Designed and deployed a PLC system migration to Ethernet/IP Remote I/O, increasing data acquisition and reducing network latency across the production line.
- Support the integration of new automated cells into the manufacturing ecosystem, performing electrical system modifications and validating safety interlocks.

Wegmann Automotive - *Murfreesboro, TN*
Steel Assembly Technician *Nov 2021 – Nov 2023*

- Optimized Cognex In-Sight vision inspection algorithms to achieve a high detection rate, ensuring zero-defect delivery to downstream processes.
- Collected and analyzed machine data to minimize scrap and downtime.

General Mills - *Murfreesboro, TN*
Machine Operator *May 2019 – Nov 2021*

- Identified and resolved mechanical and electrical equipment failures in high-speed food processing environments to maintain efficient OEE rating.
- Performed preventative and breakdown maintenance.
- Learned industry standards: KPI's, Project Management, Lean Manufacturing, and 5s.

EDUCATION

B.S Mechatronics Engineering | *Middle Tennessee State University* | Expected Fall 2026

A.A.S Mechatronics Technology | *Motlow State Community College* | May 2021

TECHNICAL SKILLS

- **PLCs/HMIs:** Siemens (Level 1 & 2), Mitsubishi, Yaskawa, Allen-Bradley/Rockwell.
- **Robotics/Vision:** Fanuc, Mitsubishi, Cognex In-Sight, Keyence.
- **Vision Systems:** Proficient with Cognex In-Sight software and brief experience with Keyence systems.
- **Networking/Protocols:** Ethernet/IP, Modbus/TCP, Profibus, BACnet.
- **Software:** AutoCAD, SolidWorks, Fusion 360, Python, C++, Linux, Windows, Virtual Machines.